

Status: 3-Hourly Solar Reconstruction for TCo1279-DART

Where it stands, the rescaling design, and whether to improve before scaling up

AWI Dark-Doldrums Project

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In one paragraph

DART saves 3-hourly TOA radiation and clouds but only monthly surface radiation (**ssrd**). We trained a gradient-boosting model on ERA5 (which has both) to predict **ssrd** from TOA transmission, cloud fractions and solar geometry, validated it honestly on ERA5 held-out months (98.6% of variance explained), then ran it once on real DART data (Jan 1950, TCo1279-DART-1950C) end to end. It ran cleanly and produced a physically sane result. The open question is not “does it run” — it does — but “does the ERA5-trained relationship between clouds/transmission and surface radiation actually match how DART’s radiation scheme behaves,” and the one test we have so far gives a mixed but interpretable answer, detailed below.

How the monthly value is actually used — and why not as a training input

Worth addressing directly: DART’s known monthly **ssrd** is used, but as an exact post-hoc correction, not as an input to the ML model. After the model predicts 3-hourly values, each grid cell’s predicted values are multiplied by a single factor so that their monthly mean exactly equals DART’s real monthly **ssrd** for that cell:

$$ssrd_{final}(t) = ssrd_{pred}(t) \times \frac{ssrd_{monthly, real}}{ssrd_{monthly, pred}}$$

Feeding the monthly value into the model itself as a feature instead was considered and is deliberately not what we did, for a concrete reason: doing so would only make the model’s *aggregate* output roughly consistent with the real monthly mean, not exactly consistent, because a gradient boosting model has no built-in mechanism to guarantee an aggregation identity — you’d typically still need a correction step afterward to close the residual gap. Since we need day-to-day totals to add up to a known, trusted number (this feeds capacity-factor calculations later), an exact, provable correction is preferable to an approximate one. Concretely: the post-hoc rescale currently used closes the gap to 0.0002 W/m² — far tighter than a model could be trusted to hit on its own, and unlike a learned feature, it’s independently checkable in one line for any month without touching the model.

There’s a genuine limitation this exposes, though, and it is the crux of the current open question:

The rescale is a single number per grid cell per month — it corrects the overall level, not the shape. If DART’s real day-to-day cloud variability drives surface radiation differently than ERA5’s does (different radiation scheme, different resolution), a monthly correction cannot fix that; it can only make the monthly total right while leaving the distribution of that total across

the month’s individual 3-hour steps still shaped by whatever the ERA5-trained model thinks is typical. This is exactly why the per-cell correction factors are the diagnostic to watch, not just an implementation detail.

What the one test month actually showed

Before any correction, the reconstructed January 1950 field averaged 204 W/m^2 against DART’s real monthly value of 191 W/m^2 — about 6.6% high, which is in line with the residual error already seen when the same model was tested on ERA5’s own held-out months, so nothing about the DART transfer looks broken on its face.

The per-cell correction factors needed to close that gap ranged from 0.56x to 5.3x — wider than anything seen within ERA5. But the distribution is reassuring in its middle: the median factor is exactly 1.00, and 90% of all cells need a correction between 0.94x and 1.36x, comparable to normal model error. The extreme values are a minority of specific locations, not a global symptom. That pattern — a well-behaved bulk with a smaller number of large local outliers — is the same shape of problem already documented on the wind side of this project, where onshore/terrain cells showed roughly double the error of open water. It’s a reasonable working hypothesis that the same physical drivers (terrain, coastline, possibly high-latitude winter snow/ice albedo) are responsible here too, but that hasn’t been confirmed yet for solar — the map exists but hasn’t been cross-checked against geography yet.

Can the model be improved before running more months?

Yes, and cheaply, before committing to the full 240 + 156 month run:

1. **Check whether the outliers are geographic first**, by overlaying the scale-factor map against terrain and coastline. If the pattern matches the known wind-side terrain effect, that’s a specific, fixable hypothesis rather than a diffuse “the model might be wrong” worry.
2. **Add surface albedo (fal) as a feature**. DART carries this monthly. It directly addresses a known blind spot of the current feature set: a clear sky over bright snow/ice and an overcast sky over dark ground can produce a similar TOA signature, which transmission and cloud fraction alone can’t always distinguish. If winter/high-latitude cells are over-represented among the outliers, this is the most likely single improvement available without retraining architecture.
3. **Run 3-4 more contrasting months** (different season, and one from the 2080C run) before deciding anything permanent. One month cannot tell us whether an outlier pattern is a persistent regional bias or a one-off. This is a half-day of compute, not a redesign.
4. **Deliberately not** doing: switching the rescale itself to use a different divisor, an additive correction, or folding the monthly value into training — none of these address a shape problem, only a level problem the current rescale already solves exactly.

Bottom line

The pipeline works end-to-end and the one real test is within expected error bounds after correction. The right next step is diagnostic, not a rebuild: confirm whether the wide correction-factor spread is geographic and systematic (fixable, likely via adding albedo) or scattered and unpredictable (would need a harder look at the feature set) — using a handful of additional test months — before launching the full two-period run.